

A CATEGORICAL DESCRIPTION OF SIMPLE BETH COMPANIONS

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Abstract

A pp expansion of a quasivariety $\mathbf{K}$ is said to be *simple* when it is of the form $\mathbf{K}[\mathcal{L}_{\mathcal{F}}]$. For instance, when $\mathbf{K}$ has the amalgamation property, all its pp expansions are simple. It is shown that the simple pp expansions of a quasivariety $\mathbf{K}$ coincide with the quasivarieties $\mathbf{M}$ for which the forgetful functor $U: \mathbf{M} \rightarrow \mathbf{K}$ is well defined and induces an isomorphism from $\mathbf{M}$ to a mono-reflective subcategory of $\mathbf{K}$. As a consequence, if a quasivariety $\mathbf{K}$ possesses a simple Beth companion $\mathbf{M}$, then $\mathbf{M}$ is the unique (up to term equivalence) quasivariety whose monomorphisms are regular that, moreover, satisfy the categorical description of simple pp expansions of $\mathbf{K}$ given above.

1. EXPANSIONS OF QUASIVARIETIES

We denote the class operators of closure under isomorphisms, subalgebras, direct products, finite direct products, and ultraproducts by $\mathbb{I}, \mathbb{S}, \mathbb{P}, \mathbb{P}^{<\omega}$, and $\mathbb{P}_u$, respectively. A class of similar algebras is said to be a *quasivariety* when it is closed under $\mathbb{I}, \mathbb{S}, \mathbb{P}$, and $\mathbb{P}_u$ or, equivalently, when it can be axiomatized by a set of *quasiequations*, that is, formulas of the form $\bigwedge \Phi \rightarrow \varphi$, where $\Phi \cup \{\varphi\}$ is a finite set of equations and $\bigwedge$ the conjunction symbol (see, e.g., [BS12, Thm. V.2.25]).

Let $\mathbf{K}$ be a quasivariety. A congruence θ of an algebra $\mathbf{A}$ is a *$\mathbf{K}$ -congruence* when $\mathbf{A}/\theta \in \mathbf{K}$. For every $X \subseteq A \times A$ there exists the least $\mathbf{K}$ -congruence of $\mathbf{A}$ containing X , which we denote by $\mathbf{Cg}_{\mathbf{K}}^{\mathbf{A}}(X)$ (see, e.g., [Gor98, Sec. 1.4.4, p. 39]). Every quasivariety can be viewed as a category whose objects are its members and whose arrows are the homomorphisms between them.

Remark 1.1. In every quasivariety, monomorphisms coincide with embeddings, i.e., injective homomorphisms (see, e.g., [McK96, p. 222]). $\square$

1.1. The basic adjunction. We denote the language of a quasivariety $\mathbf{K}$ by $\mathcal{L}_{\mathbf{K}}$. Given a quasivariety $\mathbf{K}$ and a language $\mathcal{L} \subseteq \mathcal{L}_{\mathbf{K}}$, we denote the $\mathcal{L}$ -reduct of a member $\mathbf{A}$ of $\mathbf{K}$ by $\mathbf{A}|_{\mathcal{L}}$. Given a pair of quasivarieties $\mathbf{K}$ and $\mathbf{M}$, the forgetful functor $U: \mathbf{M} \rightarrow \mathbf{K}$ is well defined if and only if $\mathcal{L}_{\mathbf{K}} \subseteq \mathcal{L}_{\mathbf{M}}$ and $\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}} \in \mathbf{K}$ for every $\mathbf{A} \in \mathbf{M}$.

Definition 1.2. Let $\mathbf{K}$ and $\mathbf{M}$ be a pair of quasivarieties. We say that $\mathbf{M}$ is an *expansion* of $\mathbf{K}$ when the forgetful functor $U: \mathbf{M} \rightarrow \mathbf{K}$ is well defined.

Let $\mathbf{K}$ be a quasivariety. We recall that for each nonempty set X the free algebra $\mathbf{T}_{\mathbf{K}}(X)$ over $\mathbf{K}$ with set free generators X belongs to $\mathbf{K}$ (see, e.g., [BS12, Thm. II.10.12]). We will often identify a term $t(x_1, \dots, x_n)$ of $\mathbf{K}$ with $x_1, \dots, x_n \in X$ with its equivalence class in $\mathbf{T}_{\mathbf{K}}(X)$.

Consider an expansion $\mathbf{M}$ of a quasivariety $\mathbf{K}$. The forgetful functor $U: \mathbf{M} \rightarrow \mathbf{K}$ has a left adjoint F which can be described as follows. With each $\mathbf{A} \in \mathbf{K}$ we associate a set of elements of pair of terms of $\mathbf{K}$ as follows:

$$\text{diag}^+(\mathbf{A}) = \{ \langle t(a_1, \dots, a_n), s(a_1, \dots, a_n) \rangle : t(x_1, \dots, x_n) \text{ and } s(x_1, \dots, x_n) \text{ are terms of } \mathbf{K}, \\ a_1, \dots, a_n \in A, \text{ and } t^{\mathbf{A}}(a_1, \dots, a_n) = s^{\mathbf{A}}(a_1, \dots, a_n) \}.$$

As $\mathbf{M}$ is an expansion of $\mathbf{K}$, we can view $\text{diag}^+(\mathbf{A})$ as a subset of $T_{\mathbf{M}}(A) \times T_{\mathbf{M}}(A)$. Then the algebra

$$F(\mathbf{A}) = \mathbf{T}_{\mathbf{M}}(A)/\theta_{\mathbf{A}}, \text{ where } \theta_{\mathbf{A}} = \text{Cg}_{\mathbf{M}}^{T_{\mathbf{M}}(A)}(\text{diag}^+(\mathbf{A}))$$

belongs to $\mathbf{M}$ because θ is an $\mathbf{M}$ -congruence of $\mathbf{T}_{\mathbf{M}}(A)$ by definition. Furthermore, for each homomorphism $h: \mathbf{A} \rightarrow \mathbf{B}$ with $\mathbf{A}, \mathbf{B} \in \mathbf{K}$ let $F(h): F(\mathbf{A}) \rightarrow F(\mathbf{B})$ be the homomorphism defined for all $a_1, \dots, a_n \in A$ and terms $t(x_1, \dots, x_n)$ of $\mathbf{M}$ as $F(h)(t(a_1, \dots, a_n)/\theta_{\mathbf{A}}) = t(h(a_1), \dots, h(a_n))/\theta_{\mathbf{B}}$. We call $F: \mathbf{K} \rightarrow \mathbf{M}$ the *free extension functor* associated with $U: \mathbf{M} \rightarrow \mathbf{K}$.

We denote the unit and the counit of the adjunction $F \dashv U$ by $\eta: \text{id}_{\mathbf{K}} \rightarrow UF$ and $\epsilon: FU \rightarrow \text{id}_{\mathbf{M}}$, respectively. For every $\mathbf{A} \in \mathbf{K}$ the map $\eta_{\mathbf{A}}: \mathbf{A} \rightarrow UF(\mathbf{A})$ is defined for every $a \in A$ as $\eta_{\mathbf{A}}(a) = a/\theta_{\mathbf{A}}$. Moreover, for every $\mathbf{B} \in \mathbf{M}$ the map $\epsilon_{\mathbf{B}}: FU(\mathbf{B}) \rightarrow \mathbf{B}$ is defined for all $b_1, \dots, b_n \in B$ and terms $t(x_1, \dots, x_n)$ of $\mathbf{M}$ as $\epsilon_{\mathbf{B}}(t(b_1, \dots, b_n)/\theta_{U(\mathbf{B})}) = t^{\mathbf{B}}(b_1, \dots, b_n)$.

1.2. Implicit operations. An *implicit operation* of a quasivariety $\mathbf{K}$ is a family of partial functions on the members of $\mathbf{K}$ that is globally preserved by homomorphisms and definable by a formula (see [CKM25, Sec. 3]). More precisely, an n -ary *operation* of $\mathbf{K}$ is a sequence $f = \langle f^{\mathbf{A}} : \mathbf{A} \in \mathbf{K} \rangle$, where each $f^{\mathbf{A}}: \text{dom}(f^{\mathbf{A}}) \rightarrow A$ is a partial n -ary function on A with domain $\text{dom}(f^{\mathbf{A}}) \subseteq A^n$ that is globally preserved by the homomorphisms between members of $\mathbf{K}$. The latter means that for every homomorphism $h: \mathbf{A} \rightarrow \mathbf{B}$ with $\mathbf{A}, \mathbf{B} \in \mathbf{K}$ and $\langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{A}})$ we have

$$\langle h(a_1), \dots, h(a_n) \rangle \in \text{dom}(f^{\mathbf{B}}) \text{ and } h(f^{\mathbf{A}}(a_1, \dots, a_n)) = f^{\mathbf{B}}(h(a_1), \dots, h(a_n)).$$

For $n \geq 1$, an n -ary operation f of $\mathbf{K}$ is said to be *implicit* when it is defined by some first order formula $\varphi(x_1, \dots, x_n, y)$, in the sense that for all $\mathbf{A} \in \mathbf{K}$ and $a_1, \dots, a_n, b \in A$,

$$\mathbf{A} \models \varphi(a_1, \dots, a_n, b) \iff \langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{A}}) \text{ and } f^{\mathbf{A}}(a_1, \dots, a_n) = b.$$

For instance, “taking inverses” is an implicit operation of the class of all monoids because it can be defined by the conjunction of equations $\varphi(x, y) = (xy \approx 1) \sqcap (yx \approx 1)$ and monoid homomorphisms preserve inverses when they exist.

Notably, implicit operations admit a description in terms of *primitive positive formulas* (for short, *pp formulas*), that is, formulas of the form $\exists x_1, \dots, x_n \varphi$, where φ is a conjunction of equations. More precisely, if f is an implicit operation of a quasivariety $\mathbf{K}$, there exist implicit operations $f_1, \dots, f_n$ of $\mathbf{K}$ definable by pp formulas such that $f^{\mathbf{A}} = f_1^{\mathbf{A}} \cup \dots \cup f_n^{\mathbf{A}}$ for every $\mathbf{A} \in \mathbf{K}$ (see [CKM25, Cor. 3.10]). Consequently, the pp definable implicit operations of $\mathbf{K}$ form the building blocks of all implicit operations of $\mathbf{K}$ and, therefore, we restrict our attention to them. The next result simplifies the task of determining whether a function can be defined by a pp formula (see [CV15, Thm. 6.3(4)]).

Theorem 1.3. *Let $\mathbf{K}$ be a class of algebras, $f \in \mathcal{L}_{\mathbf{K}}$, and $\mathcal{L} \subseteq \mathcal{L}_{\mathbf{K}}$. Then there exists a pp formula of $\mathcal{L}$ that defines f in $\mathbf{K}$ if and only if f is preserved by every homomorphism $h: \mathbf{A}|_{\mathcal{L}} \rightarrow \mathbf{B}|_{\mathcal{L}}$ with $\mathbf{A}, \mathbf{B} \in \mathbb{P}_{\mathbf{u}}\mathbb{P}^{<\omega}(\mathbf{K})$.*

In general, the implicit operations of a quasivariety $\mathbf{K}$ need not be componentwise total. Therefore, we say that an implicit operation f of $\mathbf{K}$ is *extendable* when for all $\mathbf{A} \in \mathbf{K}$ and $\langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{A}})$ there exists an algebra $\mathbf{B} \in \mathbf{K}$ extending $\mathbf{A}$ such that $\langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{B}})$. The class of extendable implicit operations of $\mathbf{K}$ will be denoted by $\text{ext}(\mathbf{K})$, and that of pp definable extendable implicit operations of $\mathbf{K}$ by $\text{ext}_{\text{pp}}(\mathbf{K})$. The next result justifies the term “extendable” (see [CKM25, Prop. 8.1 and Thm. 8.4]).

Theorem 1.4. *Let $\mathbf{K}$ be a quasivariety and $\mathbf{A} \in \mathbf{K}$. Then there exists $\mathbf{B} \in \mathbf{K}$ with $\mathbf{A} \leq \mathbf{B}$ such that $f^{\mathbf{B}}$ is total and extends $f^{\mathbf{A}}$ for each $f \in \text{ext}(\mathbf{K})$.*

The task of constructing extendable implicit operations is simplified by the following observation (see [CKM25, Cor. 3.11]).

Proposition 1.5. *Let $\mathbf{K}$ be a quasivariety and $\varphi(x_1, \dots, x_n, y)$ a pp formula. If every $\mathbf{A}$ can be extended to some $\mathbf{B} \in \mathbf{K}$ on which φ defines a total n -ary function, then φ defines an n -ary member of $\text{ext}_{\text{pp}}(\mathbf{K})$.*

In order to add a family of implicit operations $\mathcal{F} \subseteq \text{ext}_{\text{pp}}(\mathbf{K})$ to a quasivariety $\mathbf{K}$, we proceed as follows. Let $\mathcal{L}_{\mathcal{F}}$ the language obtained by adding to $\mathcal{L}_{\mathbf{K}}$ a new n -ary function symbol g_f for each n -ary $f \in \mathcal{F}$. Then we expand every member $\mathbf{A}$ of $\mathbf{K}$ in which $\{f^{\mathbf{A}} : f \in \mathcal{F}\}$ is a family of total functions to an algebra $\mathbf{A}[\mathcal{L}_{\mathcal{F}}]$ in the language $\mathcal{L}_{\mathcal{F}}$ by interpreting g_f as $f^{\mathbf{A}}$ for each $f \in \mathcal{F}$. The pp expansion of $\mathbf{K}$ induced by $\mathcal{F}$ is $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$. We will make use of the following observation (see [CKM25, Prop. 10.2]).

Proposition 1.6. *Let $\mathbf{M}$ be a pp expansion of a quasivariety $\mathbf{K}$. Then $\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}} \in \mathbf{K}$ for every $\mathbf{A} \in \mathbf{M}$.*

2. SIMPLE BETH COMPANIONS

In this note, we shall focus on the following kind of pp expansions.

Definition 2.1. A pp expansion of a quasivariety $\mathbf{K}$ is said to be *simple* when it is of the form $\mathbf{K}[\mathcal{L}_{\mathcal{F}}]$ for some $\mathcal{F} \subseteq \text{ext}_{\text{pp}}(\mathbf{K})$.

Simple pp expansions are relatively common, as witnessed by the following.

Theorem 2.2. *Every pp expansion of a quasivariety with the amalgamation property is simple.*

Another source of pp expansions derives from the following kind of implicit operations.

Definition 2.3. Let $\mathbf{K}$ be a quasivariety and $f \in \text{imp}_{\text{pp}}(\mathbf{K})$. We say that f has *unique witnesses* when it can be defined by pp formula

$$\exists z_1, \dots, z_m \varphi(z_1, \dots, z_m, x_1, \dots, x_n, y)$$

such that for all $\mathbf{A} \in \mathbf{K}$ and $\langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{A}})$ there exists a unique tuple $\langle b_1, \dots, b_m \rangle \in A^m$ satisfying

$$\mathbf{A} \models \varphi(b_1, \dots, b_m, a_1, \dots, a_n, f^{\mathbf{A}}(a_1, \dots, a_n)).$$

We denote by $\text{imp}_{\text{pp!}}(\mathbf{K})$ the subset of $\text{imp}_{\text{pp}}(\mathbf{K})$ consisting of the definable implicit operations with unique witnesses. We also let $\text{ext}_{\text{pp!}}(\mathbf{K}) = \text{ext}_{\text{pp}}(\mathbf{K}) \cap \text{imp}_{\text{pp!}}(\mathbf{K})$.

We also recall that a pp expansion $\mathbf{M}$ of a quasivariety $\mathbf{K}$ is said to be a *Beth companion* of $\mathbf{K}$ when monomorphisms are regular in $\mathbf{M}$ (see [CKM25, Rmk. 6.4 & Thm. 11.6]). Although a quasivariety $\mathbf{K}$ may lack a Beth companion (see, e.g., [CKM25, Thm. 14.17] and [CKM26, Thm. 6.1]), when it possesses one, it must be essentially unique, as we proceed to illustrate.

Let $\mathbf{M}_1$ and $\mathbf{M}_2$ be a pair of pp expansions of a quasivariety $\mathbf{K}$. For $i = 1, 2$ let T_i be the set of terms of $\mathbf{M}_i$ with variables in $\{x_n : n \in \mathbb{N}\}$. Let $\rho : \mathcal{L}_{\mathbf{M}_2} \rightarrow T_1$ be a map that preserves the arities. For each $\mathcal{L}_{\mathbf{M}_1}$ -algebra $\mathbf{A}$ let $\rho(\mathbf{A})$ be the $\mathcal{L}_{\mathbf{M}_2}$ -algebra with universe A such that $f^{\rho(\mathbf{A})} = \rho(f)^{\mathbf{A}}$ for each function symbol f in $\mathcal{L}_{\mathbf{M}_2}$. Similarly, given an arity-preserving map $\tau : \mathcal{L}_{\mathbf{M}_1} \rightarrow T_2$ and an $\mathcal{L}_{\mathbf{M}_2}$ -algebra $\mathbf{B}$, we define an $\mathcal{L}_{\mathbf{M}_1}$ -algebra $\tau(\mathbf{B})$. We say that $\mathbf{M}_1$ and $\mathbf{M}_2$ are *faithfully term equivalent relative to $\mathbf{K}$* if there exist arity-preserving maps $\tau : \mathcal{L}_{\mathbf{M}_1} \rightarrow T_2$ and $\rho : \mathcal{L}_{\mathbf{M}_2} \rightarrow T_1$ such that $\tau(f) = f(x_1, \dots, x_n)$ and $\rho(f) = f(x_1, \dots, x_n)$ for each n -ary function symbol f in $\mathcal{L}_{\mathbf{K}}$, and for all $\mathbf{A} \in \mathbf{M}_1$ and $\mathbf{B} \in \mathbf{M}_2$ we have

- (i) $\rho(\mathbf{A}) \in \mathbf{M}_2$;
- (ii) $\tau(\mathbf{B}) \in \mathbf{M}_1$;
- (iii) $\tau\rho(\mathbf{A}) = \mathbf{A}$;
- (iv) $\rho\tau(\mathbf{B}) = \mathbf{B}$.

When they exist, Beth companions are essentially unique in the following sense.

Theorem 2.4 ([CKM25, Thm. 11.7]). *All the Beth companions of a quasivariety $\mathbf{K}$ are faithfully term equivalent relative to $\mathbf{K}$.*

We will make use of the following characterization of Beth companions (see [CKM25, Thm. 11.6]).

Theorem 2.5. *The following are equivalent for a pp expansion $\mathbf{M}$ of a quasivariety $\mathbf{K}$:*

- (i) $\mathbf{M}$ is a Beth companion of $\mathbf{K}$;
- (ii) for all $\mathbf{A} \in \mathbf{M}$, $f \in \text{imp}(\mathbf{K})$, and $\langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{A} \upharpoonright \mathcal{L}_{\mathbf{K}}})$ there exists a term t of $\mathbf{M}$ such that

$$t^{\mathbf{A}}(a_1, \dots, a_n) = f^{\mathbf{A} \upharpoonright \mathcal{L}_{\mathbf{K}}}(a_1, \dots, a_n).$$

As Beth companions are pp expansions, the concept of “simplicity” applies to them as well.

Definition 2.6. A Beth companion of a quasivariety $\mathbf{K}$ is said to be *simple* when it is a simple pp expansion of $\mathbf{K}$.

Every Beth companion obtained by adding extendable pp definable implicit operations with unique witnesses is simple, as we proceed to illustrate.

Theorem 2.7. *Every Beth companion of a quasivariety $\mathbf{K}$ that is induced by a subset of $\text{ext}_{\text{pp!}}(\mathbf{K})$ is simple.*

Proof. Let $\mathcal{F} \subseteq \text{ext}_{\text{pp}}(\mathbf{K})$ and consider the corresponding pp expansion $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$. Assume that $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$ is a Beth companion of $\mathbf{K}$. We need to prove that it is simple. To this end, it suffices to show that $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}]) \subseteq \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$, for in this case $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}]) = \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$.

Consider $\mathbf{A} \in \mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$. Then $\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}} \in \mathbf{K}$ by Proposition 1.6. We will show that $f^{\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}}$ is total and coincides with $g_f^{\mathbf{A}}$ for every $f \in \mathcal{F}$ because, in this case, we would get that $\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}[\mathcal{L}_{\mathcal{F}}]$ is well defined and $\mathbf{A} = \mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}[\mathcal{L}_{\mathcal{F}}] \in \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$, as desired.

To this end, consider an n -ary $f \in \mathcal{F}$. Since $\mathcal{F} \subseteq \text{ext}_{\text{pp}}(\mathbf{K})$ by assumption, the operation f is defined by a pp formula

$$\psi = \exists z_1, \dots, z_m \varphi(z_1, \dots, z_m, x_1, \dots, x_n, y)$$

of $\mathcal{L}_{\mathbf{K}}$ such that

$$\begin{aligned} &\text{for all } \mathbf{C} \in \mathbf{K} \text{ and } \langle c_1, \dots, c_n \rangle \in \text{dom}(f^{\mathbf{C}}) \text{ there exists a unique tuple} \\ &\langle d_1, \dots, d_m \rangle \in C^m \text{ satisfying } \mathbf{C} \models \varphi(d_1, \dots, d_m, c_1, \dots, c_n, f^{\mathbf{C}}(c_1, \dots, c_n)). \end{aligned} \quad (1)$$

for all $\mathbf{C} \in \mathbf{K}$ and $\langle c_1, \dots, c_n \rangle \in \text{dom}(f^{\mathbf{C}})$ there exists a unique tuple $\langle d_1, \dots, d_m \rangle \in C^m$ satisfying

$$\mathbf{C} \models \varphi(d_1, \dots, d_m, c_1, \dots, c_n, f^{\mathbf{C}}(c_1, \dots, c_n)).$$

Consider $a_1, \dots, a_n \in A$. As $\mathbf{A} \in \mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$, there exists $\mathbf{B} \in \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$ with $\mathbf{A} \leq \mathbf{B}$. From $\mathbf{B} \in \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$ it follows that $g_f^{\mathbf{B}}$ is defined by ψ . Together with $\mathbf{A} \leq \mathbf{B}$, this guarantees the existence of a tuple $\langle b_1, \dots, b_m \rangle \in B^m$ such that

$$\mathbf{B} \models \varphi(b_1, \dots, b_m, a_1, \dots, a_n, g_f^{\mathbf{A}}(a_1, \dots, a_n)). \quad (2)$$

To conclude the proof, it will be enough to show that $b_1, \dots, b_m \in A$.

For suppose this is the case. As φ is a formula of $\mathcal{L}_{\mathbf{K}}$, $\mathbf{A} \leq \mathbf{B}$, and $a_1, \dots, a_n, g_f^{\mathbf{A}}(a_1, \dots, a_n) \in A$, the above display and the assumption that $b_1, \dots, b_m \in A$ imply

$$\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}} \models \varphi(b_1, \dots, b_m, a_1, \dots, a_n, g_f^{\mathbf{A}}(a_1, \dots, a_n)).$$

Since ψ defines f , we obtain $\langle a_1, \dots, a_n \rangle \in \text{dom}(f^{\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}})$ and $f^{\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}}(a_1, \dots, a_n) = g_f^{\mathbf{A}}(a_1, \dots, a_n)$. Hence, we conclude that $f^{\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}}$ is total and coincides with $g_f^{\mathbf{A}}$, as desired.

Therefore, it only remains to show that $b_1, \dots, b_m \in A$. Consider a positive $k \leq m$. We will show that $b_k \in A$. Consider the pp formula

$$\psi_k(x_1, \dots, x_n, y, z_k) = \exists z_1, \dots, z_{k-1}, z_{k+1}, \dots, z_m \varphi(z_1, \dots, z_m, x_1, \dots, x_n, y)$$

of $\mathcal{L}_{\mathbf{K}}$. We will show that ψ_k defines a member of $\text{imp}_{\text{pp}}(\mathbf{K})$. Since pp formulas are preserved by homomorphisms, it suffices to prove that ψ_k defines a partial function on the members of $\mathbf{K}$. To this end, consider $\mathbf{C} \in \mathbf{K}$ and $c_1, \dots, c_{n+1}, d, e \in C$ such that

$$\mathbf{C} \models \psi_k(c_1, \dots, c_{n+1}, d) \sqcap \psi_k(c_1, \dots, c_{n+1}, e).$$

By the definition of ψ_k there exist $p_1, \dots, p_{k-1}, p_{k+1}, \dots, p_m, q_1, \dots, q_{k-1}, q_{k+1}, \dots, q_m \in C$ such that

$$\mathbf{C} \models \varphi(p_1, \dots, p_{k-1}, d, p_{k+1}, \dots, p_m, c_1, \dots, c_n, c_{n+1});$$

$$\mathbf{C} \models \varphi(q_1, \dots, q_{k-1}, e, q_{k+1}, \dots, q_m, c_1, \dots, c_n, c_{n+1}).$$

Together with the assumption that ψ defines f , the above display yields $\langle c_1, \dots, c_n \rangle \in \text{dom}(f^C)$ and $c_{n+1} = f^C(c_1, \dots, c_n)$. Consequently, (1) ensures that $e = d$. Hence, we conclude that ψ_k defines a member f_k of $\text{imp}_{\text{pp}}(\mathbf{K})$.

Recall that $\mathbf{B}$ belongs to the pp expansion $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$ of $\mathbf{K}$. Then $\mathbf{B}|_{\mathcal{L}_{\mathbf{K}}} \in \mathbf{K}$ by Proposition 1.6. From (2) and the assumption that ψ_k defines f_k it follows that

$$f_k^{\mathbf{B}|_{\mathcal{L}_{\mathbf{K}}}}(a_1, \dots, a_n, g_f^{\mathbf{A}}(a_1, \dots, a_n)) = b_k.$$

As $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$ is a Beth companion of $\mathbf{K}$ by assumption, we can apply Theorem 2.5, obtaining a term $t(x_1, \dots, x_{n+1})$ of $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$ such that

$$t^{\mathbf{B}}(a_1, \dots, a_n, g_f^{\mathbf{A}}(a_1, \dots, a_n)) = b_k.$$

Since $\mathbf{A} \leq \mathbf{B}$ by assumption and $a_1, \dots, a_n, g_f^{\mathbf{A}}(a_1, \dots, a_n) \in A$, we conclude that $b_k \in A$. $\square$

3. THE MAIN RESULT

We recall that a full subcategory $\mathbf{M}$ of a category $\mathbf{K}$ is said *reflective* when the inclusion functor $i: \mathbf{M} \rightarrow \mathbf{K}$ has a left adjoint. In this case, the counit of the resulting adjunction is a natural isomorphism (see, e.g., [AHS06, Thm. 19.14(4)]). When, in addition, the unit is componentwise a monomorphism, we say that $\mathbf{M}$ is a *mono-reflective* subcategory of $\mathbf{K}$ (see, e.g., [AHS06, Def. 16.1]). Lastly, given a functor $F: \mathbf{K} \rightarrow \mathbf{M}$, we denote the direct image of $\mathbf{K}$ under F , viewed as a subcategory of $\mathbf{M}$, by $F[\mathbf{K}]$. Clearly, $F[\mathbf{K}]$ is full if and only if so is F .

Our main result is the following categorical description of simple pp expansions.

Theorem 3.1. *Let $\mathbf{M}$ be an expansion of a quasivariety $\mathbf{K}$. Then the following are equivalent:*

- (i) $\mathbf{M}$ is a simple pp expansion of $\mathbf{K}$;
- (ii) the unit of the adjunction $F \dashv U$ is componentwise a monomorphism and the counit is a natural isomorphism;
- (iii) the forgetful functor $U: \mathbf{M} \rightarrow U[\mathbf{M}]$ is an isomorphism from $\mathbf{M}$ to a mono-reflective subcategory of $\mathbf{K}$.

The following description of simple Beth companion is an immediate consequence of Theorem 3.1.

Corollary 3.2. *Let $\mathbf{M}$ be an expansion of a quasivariety $\mathbf{K}$. Then $\mathbf{M}$ is a simple Beth companion of $\mathbf{K}$ if and only if monomorphisms are regular in $\mathbf{M}$ and any of the equivalent conditions in Theorem 3.1 holds.*

We shall now prove Theorem 3.1.

Proof. (iii) $\Rightarrow$ (ii): Straightforward. (ii) $\Rightarrow$ (iii): We begin by showing that the forgetful functor $U: \mathbf{M} \rightarrow U[\mathbf{M}]$ is an isomorphism. As it is always bijective on objects, it suffices to show that it is fully faithful. The latter holds because the counit of the adjunction $F \dashv U$ is a natural isomorphism by assumption and this always guarantees that the full faithfulness of the right adjoint (see, e.g., [AHS06, Thm. 19.14(4)]).

Next, we prove that $U[\mathbf{M}]$ is a mono-reflective subcategory of $\mathbf{K}$. First, $U[\mathbf{M}]$ is a full subcategory of $\mathbf{K}$ because $U: \mathbf{M} \rightarrow \mathbf{K}$ is full. As the unit of the adjunction $F \dashv U$ is

componentwise a monomorphism and $U : \mathbf{M} \rightarrow U[\mathbf{M}]$ is an isomorphism, we deduce that $U[\mathbf{M}]$ is a mono-reflective subcategory of $\mathbf{K}$.

(ii) $\Rightarrow$ (i): We begin with the following observation.

Claim 3.3. *For every n -ary $f \in \mathcal{L}_{\mathbf{M}}$ there exists a pp formula $\varphi_f(x_1, \dots, x_n, y)$ of $\mathcal{L}_{\mathbf{K}}$ such that for all $\mathbf{A} \in \mathbf{M}$ and $a_1, \dots, a_n, b \in A$,*

$$f^{\mathbf{A}}(a_1, \dots, a_n) = b \iff \mathbf{A} \models \varphi_f(a_1, \dots, a_n, b).$$

Proof of the Claim. As $\mathbf{M}$ is a quasivariety, it is closed under $\mathbb{P}$ and $\mathbb{P}_{\mathbf{u}}$. Therefore, in view of Theorem 1.3, it suffices to show that $U : \mathbf{M} \rightarrow \mathbf{K}$ is full, which can be shown as in the proof of the implication (ii) $\Rightarrow$ (iii). $\square$

Next, we verify the following.

Claim 3.4. *For each $f \in \mathcal{L}_{\mathbf{M}}$ the formula φ_f defines a member f^* of $\text{ext}_{\text{pp}}(\mathbf{K})$.*

Proof of the Claim. From Claim 3.3 it follows that the pp formula $\varphi_f(x_1, \dots, x_n, y)$ defines a total n -ary function on each member of $\mathbf{M}$. As φ_f is a formula of $\mathcal{L}_{\mathbf{K}}$, it also defines a total n -ary function on each member of $\{U(\mathbf{B}) : \mathbf{B} \in \mathbf{M}\}$. Consequently, in view of Proposition 1.5, it suffices to show that every member $\mathbf{A}$ of $\mathbf{K}$ embeds into an algebra of the form $U(\mathbf{B})$ with $\mathbf{B} \in \mathbf{M}$. To this end, consider $\mathbf{A} \in \mathbf{K}$. By assumption the map $\epsilon_{\mathbf{A}} : \mathbf{A} \rightarrow UF(\mathbf{A})$ is a monomorphism with $F(\mathbf{A}) \in \mathbf{M}$. By Remark 1.1 we obtain that $\mathbf{A}$ embeds into $UF(\mathbf{A})$. Thus, taking $\mathbf{B} = F(\mathbf{A})$, we are done. $\square$

By Claim 3.4 the set $\mathcal{F} = \{f^* : f \in \mathcal{L}_{\mathbf{M}} - \mathcal{L}_{\mathbf{K}}\}$ induces a pp expansion $\mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$ of $\mathbf{K}$. To conclude the proof, it only remains to show that this pp expansion is simple and coincides with $\mathbf{M}$. The next claim establishes both facts at once.

Claim 3.5. *We have $\mathbf{M} = \mathbf{K}[\mathcal{L}_{\mathcal{F}}] = \mathbb{S}(\mathbf{K}[\mathcal{L}_{\mathcal{F}}])$.*

Proof of the Claim. As $\mathbf{M}$ is closed under $\mathbb{S}$ because it is a quasivariety, it suffices to show that $\mathbf{M} = \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$. We begin with the inclusion from left to right. Consider $\mathbf{A} \in \mathbf{M}$. As $U(\mathbf{A}) \in \mathbf{K}$, it suffices to show that $U(\mathbf{A})[\mathcal{L}_{\mathcal{F}}]$ is defined and coincides with $\mathbf{A}$ or, equivalently, that $f^{*U(\mathbf{A})}$ is total and coincides with $f^{\mathbf{A}}$ for each $f \in \mathcal{L}_{\mathbf{M}} - \mathcal{L}_{\mathbf{K}}$. To this end, consider an n -ary $f \in \mathcal{L}_{\mathbf{M}} - \mathcal{L}_{\mathbf{K}}$ and $a_1, \dots, a_n \in A$. By Claim 3.3 we have $\mathbf{A} \models \varphi_f(a_1, \dots, a_n, f^{\mathbf{A}}(a_1, \dots, a_n))$. Since φ_f is a formula in $\mathcal{L}_{\mathbf{K}}$, this yields $U(\mathbf{A}) \models \varphi_f(a_1, \dots, a_n, f^{\mathbf{A}}(a_1, \dots, a_n))$. As φ_f defines f^* by Claim 3.4, we conclude that $\langle a_1, \dots, a_n \rangle \in f^{*U(\mathbf{A})}$ and $f^{*U(\mathbf{A})}(a_1, \dots, a_n) = f^{\mathbf{A}}(a_1, \dots, a_n)$. Hence, $f^{*U(\mathbf{A})}$ is total and coincides with $f^{\mathbf{A}}$.

Next, we prove the inclusion $\mathbf{K}[\mathcal{L}_{\mathcal{F}}] \subseteq \mathbf{M}$. Consider $\mathbf{A} \in \mathbf{K}[\mathcal{L}_{\mathcal{F}}]$. By the definition of $\mathbf{K}[\mathcal{L}_{\mathcal{F}}]$ we have $\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}} \in \mathbf{K}$ and $\mathbf{A} = \mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}[\mathcal{L}_{\mathcal{F}}]$. It will be enough to prove that the map $h : \mathbf{A} \rightarrow F(\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}})$ defined for every $a \in A$ as $h(a) = \eta_{\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}}(a)$ is an embedding. For suppose this is the case. Then $\mathbf{A} \in \mathbb{IS}(F(\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}))$. Since $F(\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}) \in \mathbf{M}$ and $\mathbf{M}$ is closed under $\mathbb{IS}$ because it is a quasivariety, we conclude that $\mathbf{A} \in \mathbf{M}$, as desired.

Recall that $\eta_{\mathbf{A}|_{\mathcal{L}_{\mathbf{K}}}}$ is by assumption a monomorphism (i.e., an embedding by Remark 1.1). The definition of h guarantees that it is a well-defined injective map that preserves the operations in $\mathcal{L}_{\mathbf{K}}$. It only remains to prove that h preserves the operations in $\mathcal{L}_{\mathbf{M}} - \mathcal{L}_{\mathbf{K}}$ as

well. Consider an n -ary $f \in \mathcal{L}_M - \mathcal{L}_K$ and $a_1, \dots, a_n \in A$. Since $\mathbf{A} = \mathbf{A} \upharpoonright_{\mathcal{L}_K}[\mathcal{L}_F]$ and f^* is defined by φ_f by Claim 3.4 we have

$$\mathbf{A} \upharpoonright_{\mathcal{L}_K} \models \varphi_f(a_1, \dots, a_n, f^{*\mathbf{A} \upharpoonright_{\mathcal{L}_K}}(a_1, \dots, a_n)).$$

As φ_f is a pp formula in $\mathcal{L}_K$ and h preserves the operations of $\mathcal{L}_K$, we obtain that h preserves φ . Together with the above display, this implies

$$F(\mathbf{A} \upharpoonright_{\mathcal{L}_K}) \models \varphi_f(h(a_1), \dots, h(a_n), h(f^{*\mathbf{A} \upharpoonright_{\mathcal{L}_K}}(a_1, \dots, a_n))).$$

By Claim 3.3 and $F(\mathbf{A} \upharpoonright_{\mathcal{L}_K}) \in \mathbf{M}$ we conclude that

$$h(f^{*\mathbf{A} \upharpoonright_{\mathcal{L}_K}}(a_1, \dots, a_n)) = f^{F(\mathbf{A} \upharpoonright_{\mathcal{L}_K})}(h(a_1), \dots, h(a_n)).$$

Hence, $h: \mathbf{A} \rightarrow F(\mathbf{A} \upharpoonright_{\mathcal{L}_K})$ is an embedding, as desired. $\square$

(i) $\Rightarrow$ (ii): Let $\mathbf{M}$ be a simple pp expansion of $\mathbf{K}$ of the form $\mathbf{K}[\mathcal{L}_F]$ for some $\mathcal{F} \subseteq \text{ext}_{\text{pp}}(\mathbf{K})$. We begin by showing that each component of the unit η is a monomorphism. Consider $\mathbf{A} \in \mathbf{K}$. Since $\mathcal{F} \subseteq \text{ext}_{\text{pp}}(\mathbf{K})$, there exists $\mathbf{B} \in \mathbf{K}[\mathcal{L}_F]$ for which the inclusion map $i: \mathbf{A} \rightarrow U(\mathbf{B})$ is a well-defined embedding. As $F \dashv U$, there exists a homomorphism $g: F(\mathbf{A}) \rightarrow \mathbf{B}$ such that $U(g) \circ \eta_{\mathbf{A}} = i$ (see, e.g., [AHS06, Prop. 19.7(2)]). Since i is an embedding, so is $\eta_{\mathbf{A}}$. Consequently, $\eta_{\mathbf{A}}$ is a monomorphism by Remark 1.1.

Next, we prove that the counit ϵ is a natural isomorphism. To this end, it suffices to show that $U: \mathbf{M} \rightarrow \mathbf{K}$ is fully faithful (see, e.g., [AHS06, Thm. 19.14(4)]). As the forgetful functor is always faithful, it only remains to show that it is full. Consider $\mathbf{A}, \mathbf{B} \in \mathbf{M} = \mathbf{K}[\mathcal{L}_F]$ and a homomorphism $h: U(\mathbf{A}) \rightarrow U(\mathbf{B})$. From [CKM25, Prop. 9.5] it follows that h is also a homomorphism from $\mathbf{A}$ to $\mathbf{B}$. Hence, U is full, as desired. $\square$

We close this note by observing that the property of “being simple” is preserved by faithful term equivalences between pp expansions.

Corollary 3.6. *Let $\mathbf{K}$ be a quasivariety and $\mathbf{M}_1, \mathbf{M}_2$ a pair of pp expansions of $\mathbf{K}$ that are faithfully term equivalent relative to $\mathbf{K}$. The $\mathbf{M}_1$ is simple if and only if so is $\mathbf{M}_2$.*

Proof. Let $\tau: \mathcal{L}_{\mathbf{M}_1} \rightarrow T_2$ and $\rho: \mathcal{L}_{\mathbf{M}_2} \rightarrow T_1$ witness the faithful term equivalence, and assume that $\mathbf{M}_1$ is a simple pp expansion. Then Theorem 3.1 implies that the forgetful functor $U_1: \mathbf{M}_1 \rightarrow \mathbf{K}$ restricts to an isomorphism from $\mathbf{M}_1$ to a mono-reflective subcategory $U_1[\mathbf{M}_1]$ of $\mathbf{K}$. Let $U_2: \mathbf{M}_2 \rightarrow \mathbf{K}$ be the forgetful functor. Let also $\tau: \mathbf{M}_2 \rightarrow \mathbf{M}_1$ be the functor that maps $\mathbf{A} \in \mathbf{M}_2$ to $\tau(\mathbf{A}) \in \mathbf{M}_1$ and is the identity on morphisms. Similarly, define $\rho: \mathbf{M}_1 \rightarrow \mathbf{M}_2$ as the functor that maps $\mathbf{A} \in \mathbf{M}_1$ to $\rho(\mathbf{A}) \in \mathbf{M}_2$ and is the identity on morphisms. Since τ and ρ witness a term equivalence relative to $\mathbf{K}$, we have that these two functors are isomorphisms of categories that are inverses of each other and satisfy $U_1 \circ \tau = U_2$. It follows that $U_1[\mathbf{M}_1] = U_2[\mathbf{M}_2]$, and so $U_2[\mathbf{M}_2]$ is a mono-reflective subcategory of $\mathbf{K}$. Moreover, as $\tau: \mathbf{M}_2 \rightarrow \mathbf{M}_1$ and $U_1: \mathbf{M}_1 \rightarrow U_1[\mathbf{M}_1]$ are isomorphisms and $U_1 \circ \tau = U_2$ (the latter because the term equivalence witnessed by τ and ρ is faithful), we obtain that $U_2: \mathbf{M}_2 \rightarrow U_2[\mathbf{M}_2]$ is an isomorphism as well. Therefore, Theorem 3.1 yields that $\mathbf{M}_2$ is a simple pp expansion of $\mathbf{K}$. $\square$

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